

Customer Churn Prediction & Analysis

In The Telco Industry

DEPI Graduation Project · CAI4_AIS4_G2 · Group 1

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Agenda

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Introduction & Overview

Problem, dataset, objectives

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Milestones 1 & 2 — Exploratory and Advanced Data Analysis

EDA, feature engineering, statistics

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Milestone 3 — Machine Learning

5 models, tuning, winner selection

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Streamlit app, MLOps, GitHub

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Project Report and Project Presentation

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Conclusions, Recommendations and Action Plan

Insights, recommendations, action plan



1

Introduction & Overview

Presenter: Mohamed Ibrahim

Customer Churn & Project Objective

Definition

Customer churn (attrition) refers to customers who stop doing business with a company. In telecom, it means subscribers cancelling or switching to a competitor.

Churn Rate Formula

$$\text{Churn Rate} = \frac{\text{\# Churned Customers Over a Given Period}}{\text{Total \# Customers at Start of The Period}} \times 100$$

Project Objective

Building a machine learning model – utilizing data science techniques such as data collection, exploration, feature engineering, machine learning, etc. – to predict customer churn.

Why It Matters

5x

more costly to acquire
a new customer

27%

churn rate in our
dataset

\$\$\$

direct revenue
loss per churner

Customer Churn Dataset Overview

7,043

Customer Records

21

Features

27%

Churn Rate

Kaggle

Data Source

Demographics

Gender, Age, Senior Citizen, Partner, Dependents

Services

Phone, Internet, Streaming, Tech Support, Security

Account

Contract Type, Payment Method, Billing, Tenure

Target

Churn (Yes / No) — Binary Classification



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Exploratory & Advanced Data Analysis

Presenter: Ahmed Esmail

Milestone 1 — Data Collection, Exploration & Preprocessing

Exploratory Data Analysis (EDA)

Missing value and duplicate analysis
Numerical and categorical analysis
Distribution analysis

Data Cleaning

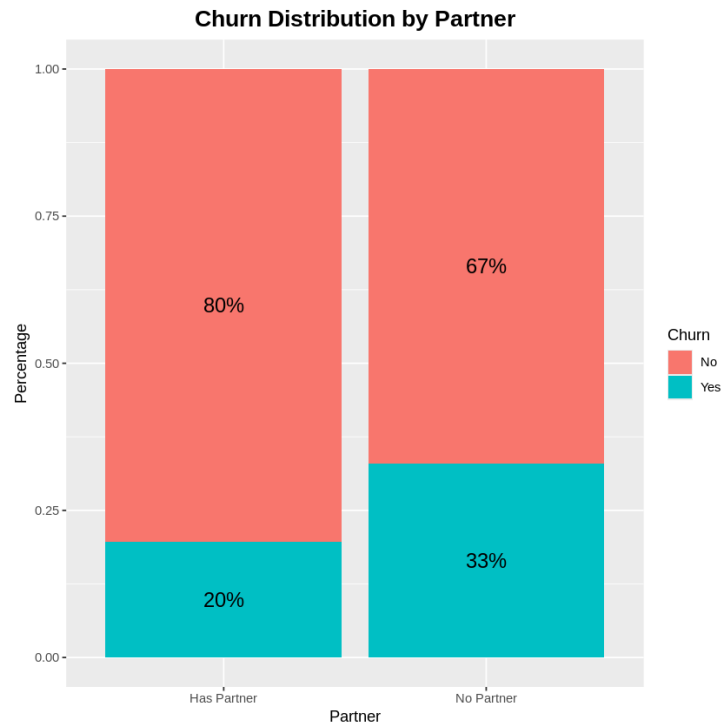
Handled missing values, outliers
Standardized inconsistent categories
Converted some datatypes into numerical formats

Visualizations

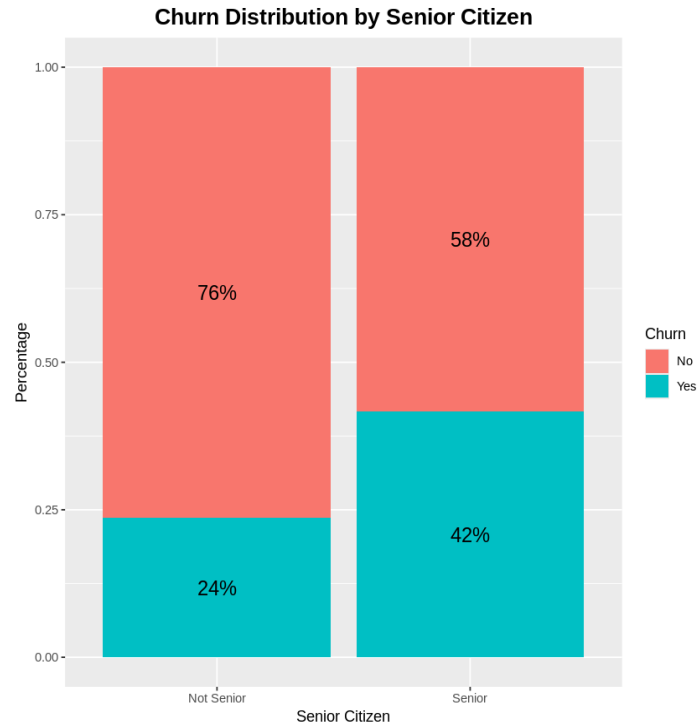
Bar Charts, Pie Charts
Histograms
Boxplots

Milestone 1 — Data Collection, Exploration & Preprocessing

Partner Analysis

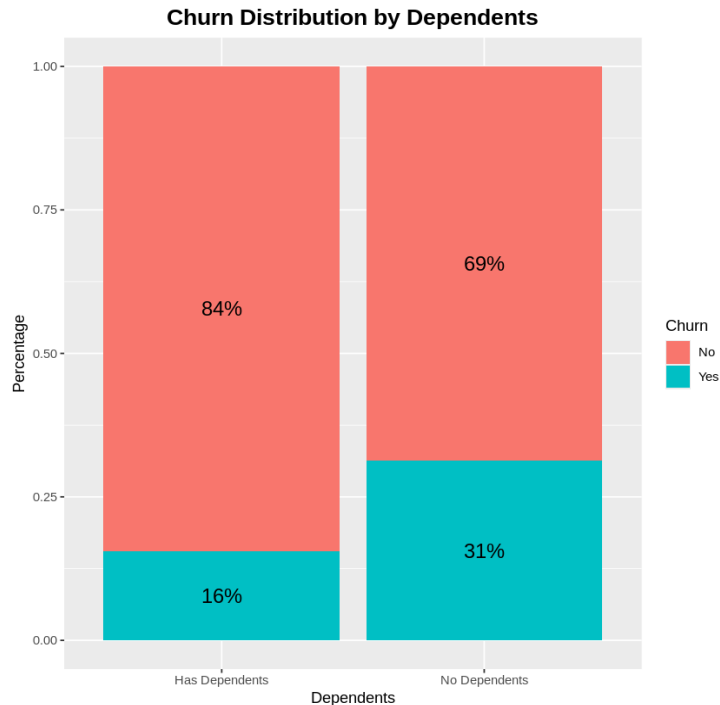


Senior Citizen Analysis

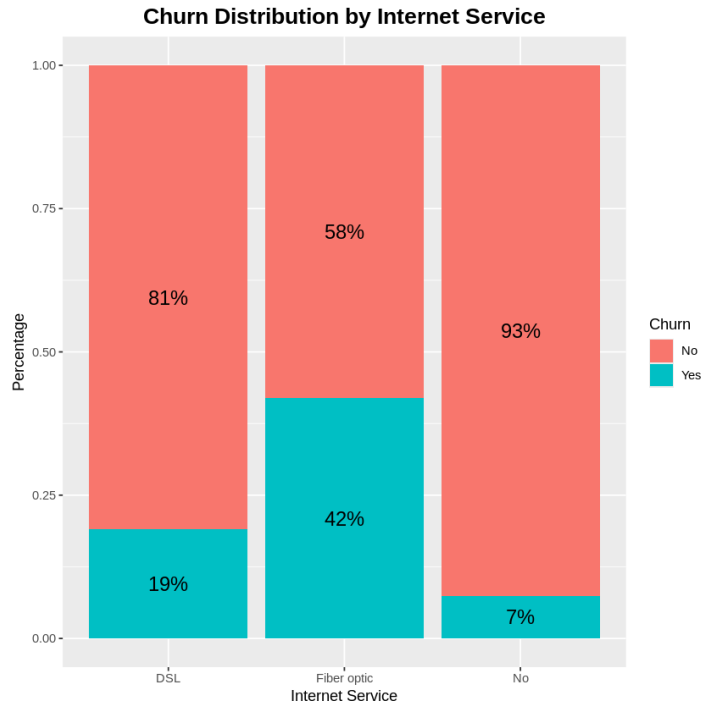


Milestone 1 — Data Collection, Exploration & Preprocessing

Dependents Analysis

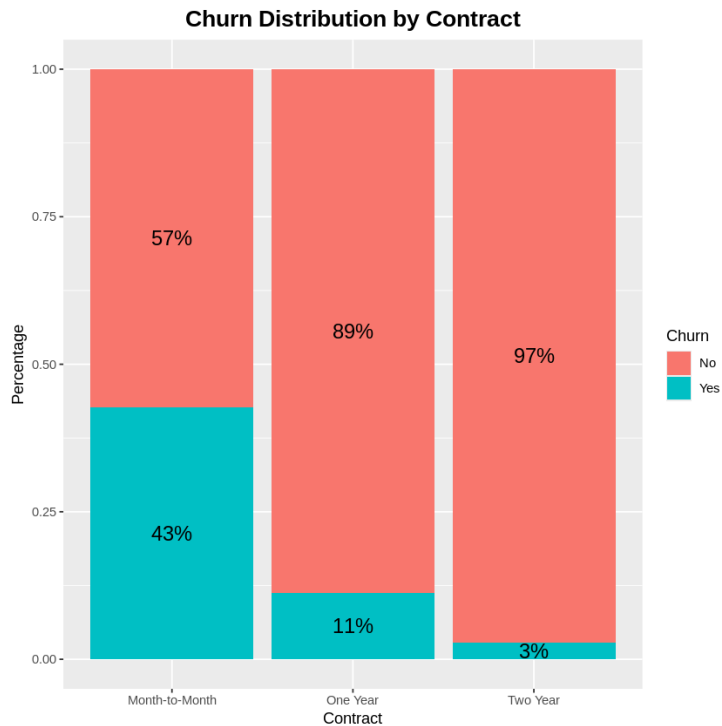


Internet Service Analysis

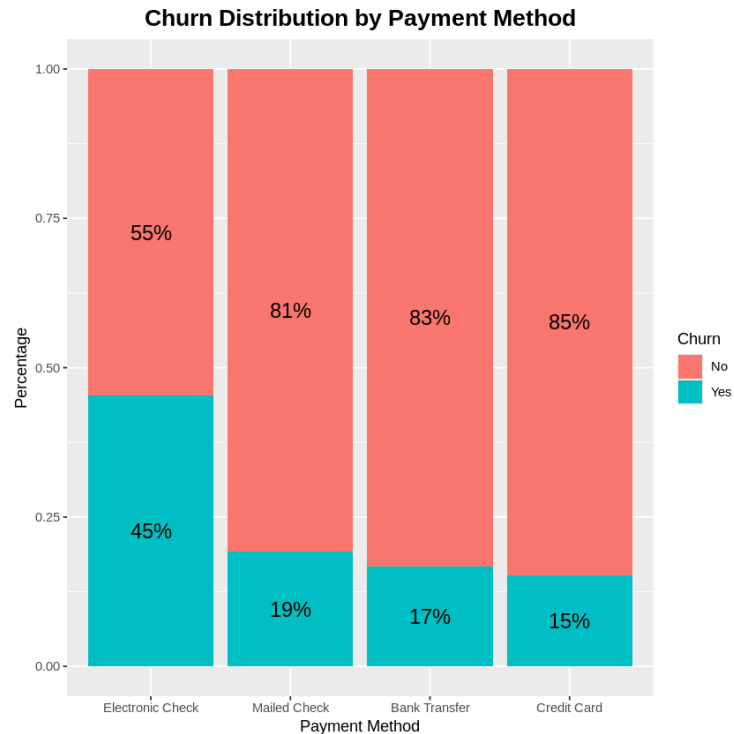


Milestone 1 — Data Collection, Exploration & Preprocessing

Contract Analysis

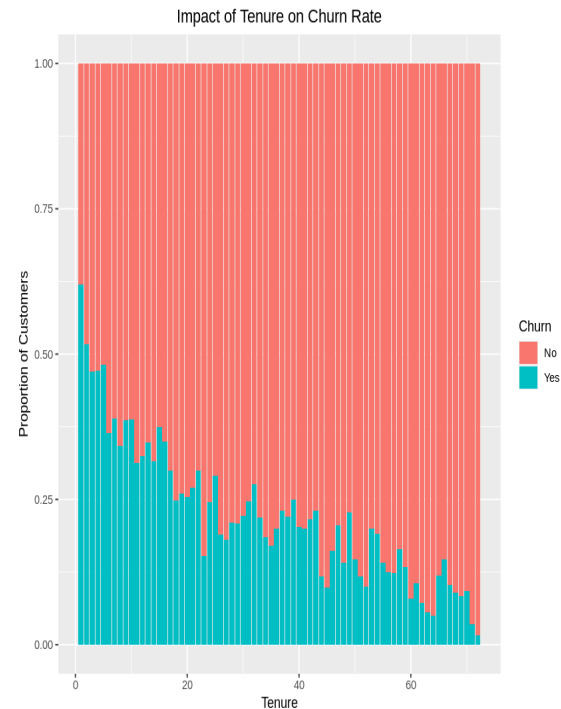
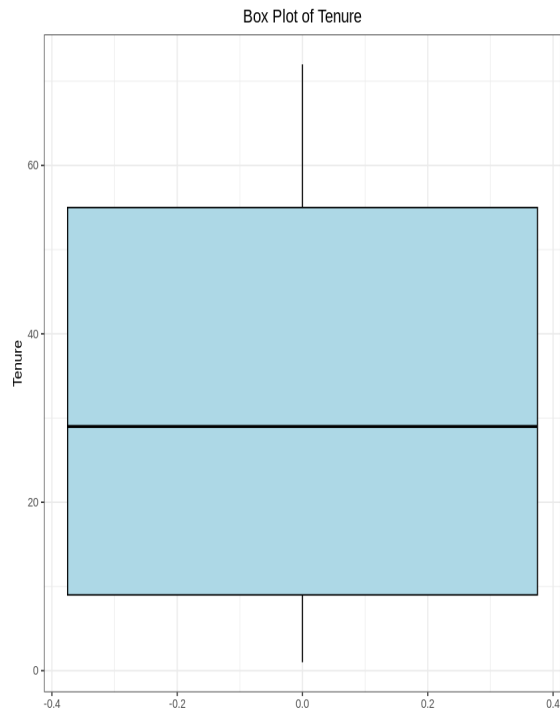
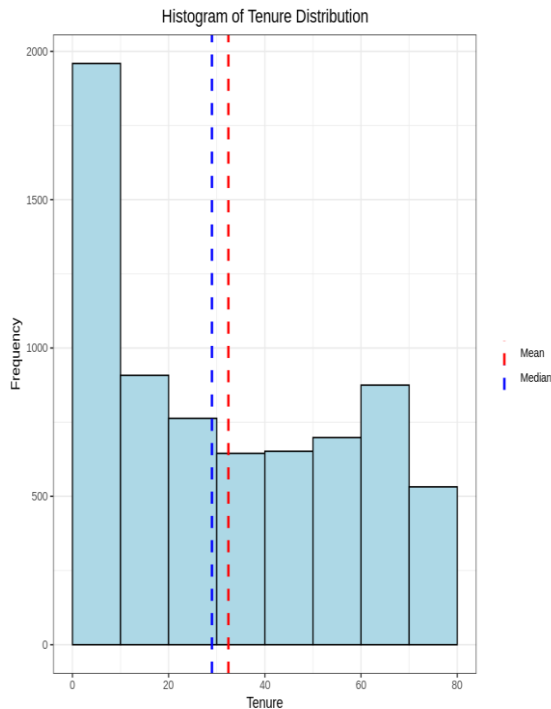


Payment Method Analysis



Milestone 1 — Data Collection, Exploration & Preprocessing

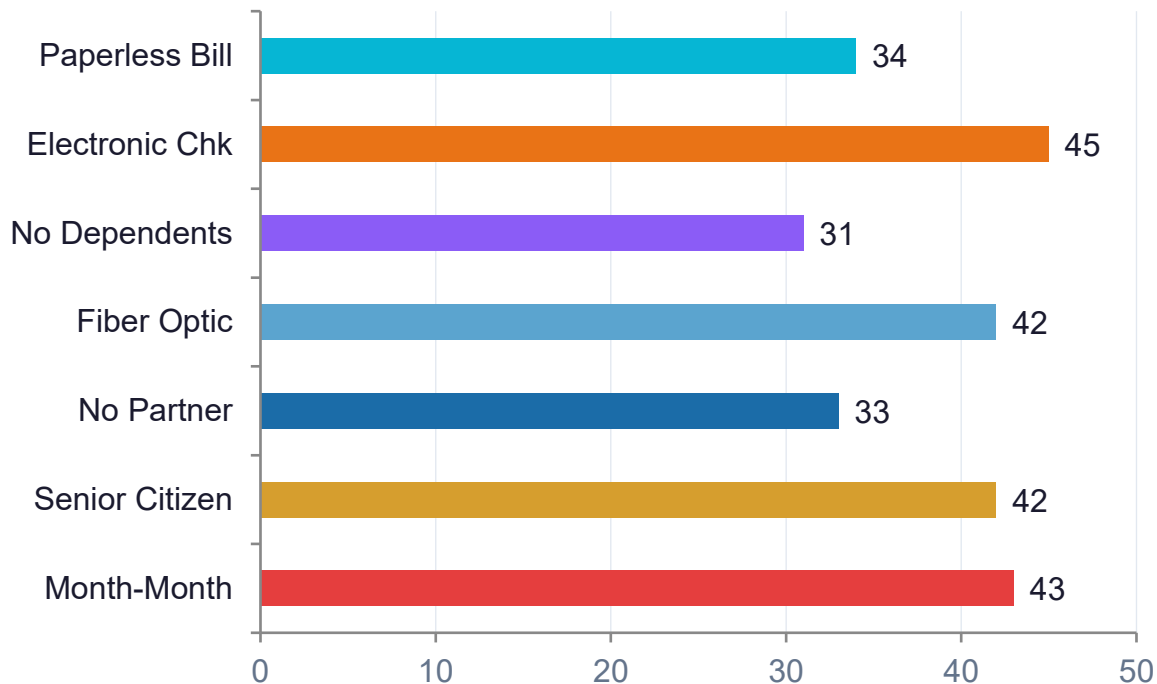
Tenure Analysis



Milestone 1 — Churn Distribution Analysis

27% overall churn rate across 7,043 customer records · Key demographic and service-level patterns identified

Churn Rate by Feature Category (%)



45%

Electronic check
churn rate (highest)



43%

Month-to-month
contract churn



3%

Two-year contract
churn (lowest)

Milestone 2 — Statistical Tests & Feature Engineering

Statistical Tests

T-Tests

Variables: Tenure, Monthly Charges, Total Charges

✓ Significant difference between churned & retained customers

Chi-Squared

Variables: Contract, Internet Service, Payment Method, Tech Support, Online Security

✓ Strong statistical relationship with churn

Engineered Features



Charge Per Service, Average Monthly Spend, Total Services,



Long-Term Customer Flag, Tenure Groups

Data Preparation for ML

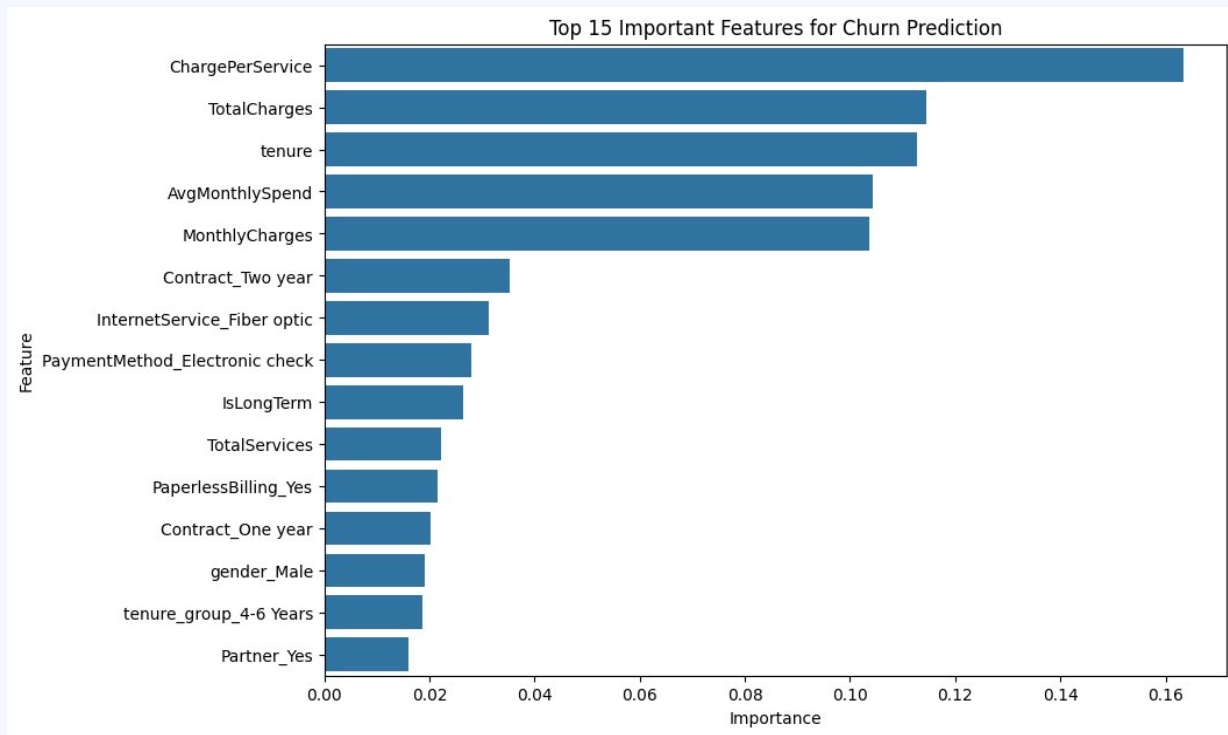


One-hot encoding for categorical variables
MinMaxScaler normalization



Feature Selection
Random Forest, RFE

Milestone 2 — Top 15 Features for Churn Prediction



Key Takeaways

- ChargePerService is the strongest predictor
- Tenure & charges dominate top 5
- Contract type & fiber optic are top categorical features
- Electronic check payment signals high risk

15 features selected for ML



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Machine Learning Model Development

Presenter: Amr El-Azab

Milestone 3 — Machine Learning

7,043

Customers Analyzed

~27%

Churn Rate

5

Models Tested & Compared

Predicting Customer Churn

Spotting customers who are likely to leave – early enough to keep them

Milestone 3 — Machine Learning

What We Did:

A clear, four-step process to build a reliable churn predictor.

1. Prepared the Data



Cleaned 7,043 customer records described by 15 signals — tenure, charges, contract type, services and more.

2. Balanced the Picture



Churners were the minority, so we balanced the training data (SMOTE) so the model learns to spot them, not ignore them.

3. Trained & Compared 5 Models



Tested five different approaches side by side on data the models had never seen, using fair, like-for-like metrics.

4. Validated & Fine-Tuned



Confirmed results were stable (5-fold cross-validation) and tuned the strongest models for the best practical performance.

Milestone 3 — Results & Recommendations

Gradient Boosting was selected as the best model for retention.

Recommendation

Deploy Gradient Boosting to score customers by churn risk and flag the highest-risk ones for proactive retention offers. The model catches roughly 4 in 5 likely leavers; the few false alarms cost only an offer, while each missed churner is a lost customer.

✓ **Deploy Gradient Boosting**



79%

of customers who leave are now correctly flagged



0.84

ROC-AUC — strong at ranking customers by risk



Gradient Boosting

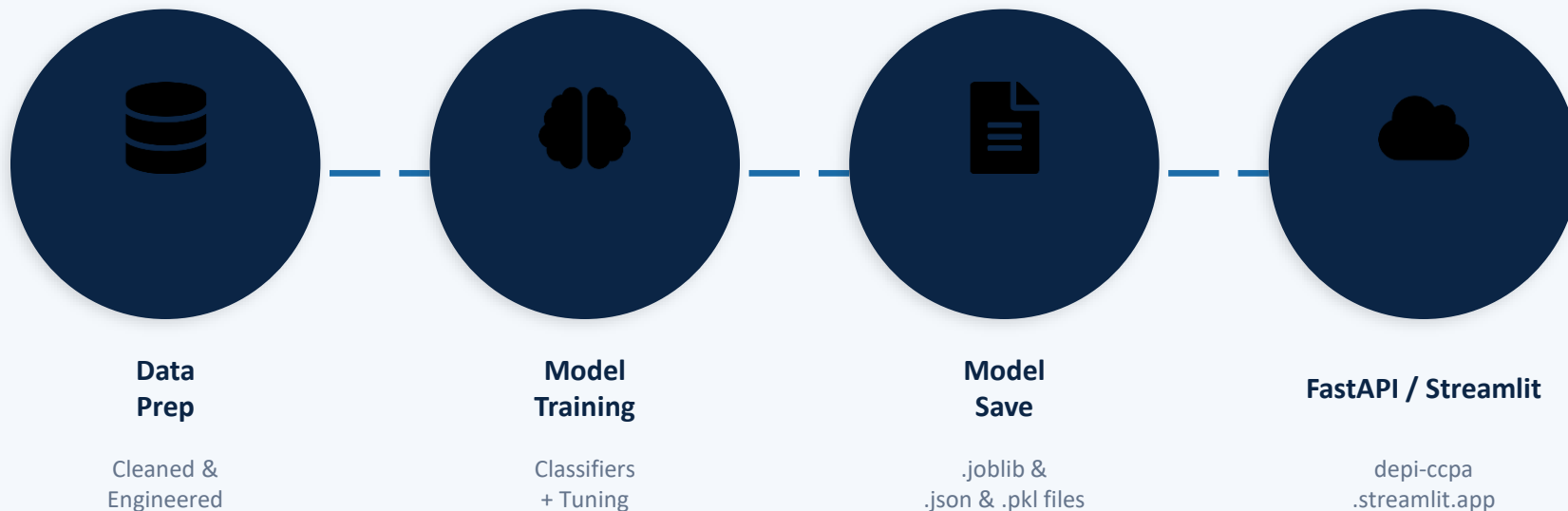
the winning model, chosen on balanced performance

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Deployment & MLOps

Presenter: Mohamed Ibrahim Rady

Milestone 4 — MLOps Pipeline & Deployment



Tools & Infrastructure

→ GitHub Repository ([Eng-MIR/DEPI4-CCPA](#))

→ Model versioning via MLflow

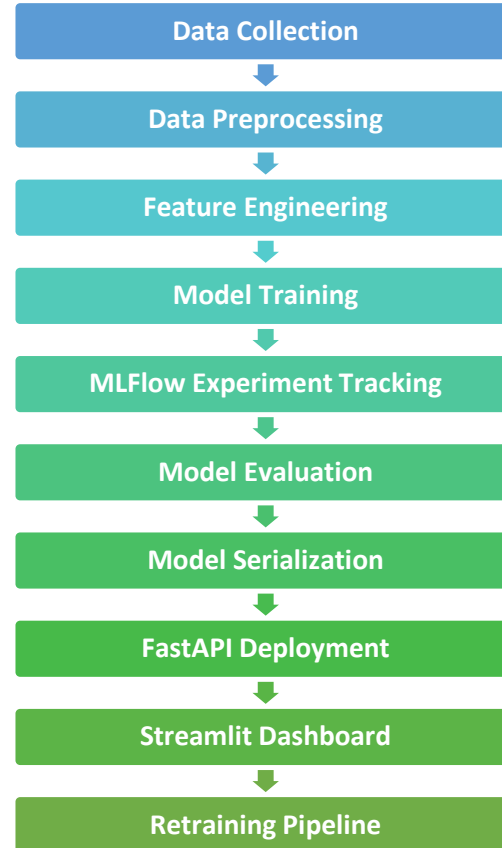
→ Streamlit Cloud deployment — live at [depi4-ccpa.streamlit.app](#)

→ 4 team members as GitHub contributors

Milestone 4 — MLOps Pipeline & Deployment

MLOps Pipeline Architecture

The implemented MLOps workflow includes:



Milestone 4 — MLOps Pipeline & Deployment

MLFlow Integration

Used MLflow for:

- Experiment tracking
- Parameter logging
- Metric tracking
- Model versioning

Saved Artifacts

- Trained model
- Scaler
- Evaluation results

Benefits

- Reproducibility
- Traceability
- Easy model comparison

Milestone 4 — Deployment, UI & Retraining Strategy

FastAPI Deployment

Model deployed using FastAPI as REST API.

API Features

- Real-time predictions
- JSON request/response
- Swagger UI testing

Streamlit Dashboard

Built interactive dashboard using Streamlit.

Dashboard Features

- User-friendly inputs
- Churn probability prediction
- Real-time results

Retraining Strategy

Retrain model

- Performance decreases
- New data becomes available

Future Improvements

- Cloud deployment
- Docker containerization
- CI/CD automation
- Automated retraining pipeline

Milestone 4 — Live Demo: Customer Churn Prediction App

The app interface is titled "Customer Churn Prediction Application". It features a sidebar on the left with "Customer Information" and a main area on the right for "Churn Prediction Probability".

Customer Information (Left Sidebar):

- Charger per Service (\$): Slider from 0 to 250, set at 250.
- Total Charges (\$): Slider from 0 to 5000, set at 5000.
- Tenure (months): Slider from 0 to 32, set at 32.
- Monthly Charges (\$): Slider from 0 to 70, set at 70.
- Average Monthly Spend (\$): Slider from 0 to 50, set at 50.
- Contract Type: Dropdown menu, set to "Month-to-month".
- Internet Service: Dropdown menu, set to "Fiber optic".
- Payment Method: Dropdown menu, set to "Electronic check".

Churn Prediction Probability (Main Area):

This app predicts Customer Churn based on Top features. Adjust the features beside and see the prediction!

High churn risk — consider immediate retention actions. (Red alert)

Low churn risk. (Green alert)

App Features

- ➔ Real-time churn prediction using Gradient Boosting model
- ➔ Interactive sliders for charges, tenure & monthly spend
- ➔ Dropdown selectors: contract type, internet, payment method
- ⚠️ Red alert — High churn risk: immediate retention action needed
- ✅ Green alert — Low churn risk: customer is stable
- ☁️ Live URL: depi4-ccpa.streamlit.app

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Documentation

Presenter: Mohamed Abdul Aziz

Milestone 5 — Project Deliverables Summary



DEPI Graduation Project

Customer Churn Analysis in Telco Industry

ROUND : CAI4_AIS4_G2

Group : 1

Team Members:

1. Mohamed Ibrahim Radwan
2. Ahmed Esmail
3. Amr el-Azab
4. Mohamed Abdul Aziz Aly

DEPI Graduation Project

Customer Churn Analysis

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Milestone 5 — Project Deliverables Summary

M1 — Data Collection & EDA

- EDA Report
- Interactive Visualizations
- Cleaned Dataset

M2 — Advanced Analysis

- Statistical Analysis Report
- Enhanced Visualizations
- Feature Engineering Summary

M5 — Documentation

- Final Project Report
- Final Presentation

M3 — ML Development

- Model Evaluation Report
- Model Code (Python)
- Final Model (Gradient Boosting)

M4 — Deployment

- Live Streamlit App
- GitHub Repository
- Model Metadata (.joblib + .json)

DEPI Graduation Project

Customer Churn Analysis

Attachments:

1. M1- Telecom dataset.csv
2. M1-2- Milestone 1 & 2_Customer_Churn_Analysis.ipynb
3. M1- final_cleaned_encoded_normalized_customer_churn.csv
4. M2- feature_importance_ranking.csv
5. M1- final_top15_selected_features_customer_churn.csv
6. All Project Visualizations.pdf
7. M3- Modeling.ipynb
8. M3- best_churn_model.joblib
9. M3- best_churn_model.info.json
10. M4- FastAPI_app.py
11. M4- Milestone4_FastAPI.ipynb
12. M4- streamlit_application.py
13. Project Presentation.pptx
14. Project Report.pdf



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Conclusions, Recommendations & Action Plan

Presenter: Mohamed Abdul Aziz

Key Conclusions from the Analysis



Contract Type is #1 Churn Driver

Month-to-month customers churn 14× more than two-year contract holders (43% vs. 3%)



Tenure is a Critical Retention Signal

Churn drops sharply after 12 months — early engagement is the most impactful intervention



Fiber Optic Customers Are High-Risk

Despite being the largest group (44%), they churn at 42% — suggesting service quality issues



Value-Added Services Reduce Churn

Customers with online security, tech support, and backup have significantly lower churn rates



Gradient Boosting is the Best Model

AUC = 0.839, Recall = 80.8% — catches 4 in 5 likely churners before they leave



Proactive Retention is Viable

ML model deployed live — enables real-time scoring and early retention interventions

Recommendations Summary

#	Finding	Recommendation
1	Month-to-month contracts have 43% churn	Incentivize migration to longer-term contracts
2	Fiber optic churn = 42%	Improve service quality; targeted retention offers
3	Electronic check churn = 45%	Incentivize switch to auto-pay or credit card
4	partner/dependents = lower churn	Family and partnership bundled service discounts (for retention)
5	Senior citizens churn at 42%	Dedicated senior support and simplified plans
6	No value-added services = higher churn	Promote security, backup, and tech support bundles
7	Short tenure customers churn more	Improve onboarding; offer first-year loyalty bonus
8	Paperless billing users churn more (34%)	Digital-first engagement and personalized comms

Action Plan — Reducing Customer Churn

1

Improve Satisfaction & Engagement

Regular surveys, faster support response, reduce complaint resolution time

2

Optimize Pricing & Contracts

Discounts for long-term commitments, flexible contract options, loyalty programs

3

Proactive Retention Program

Target high-risk customers flagged by ML model with exclusive retention offers

4

Enhance New Customer Onboarding

Welcome calls, tutorials, early engagement — reduce first-year churn

5

Improve Fiber Optic Service Quality

Address reliability issues, invest in infrastructure, premium support tiers

6

Leverage Predictive Analytics

Continuous monitoring via dashboards, retrain model quarterly on new data

Thank You

We welcome your questions

→ GitHub: github.com/Eng-MIR/DEPI4-CCPA

→ Streamlit App: depi4-ccpa.streamlit.app

→ Dataset: Kaggle — Telco Customer Churn

Team: Mohamed Ibrahim · Ahmed Esmail · Amr El-Azab · Mohamed Abdul Aziz

BACKUP SLIDES

Available on request during Q&A

Future Improvements & Retraining Strategy

1

XGBoost / LightGBM

Test more powerful gradient boosting variants for higher recall

2

Deep Learning

Explore neural networks for complex feature interaction capture

3

Real-Time Data Pipeline

Connect live CRM data feeds for continuous model scoring

4

Explainability (SHAP)

Add SHAP values to explain individual predictions to retention teams

5

Quarterly Retraining

Automated pipeline: monitor data drift → trigger retraining → redeploy

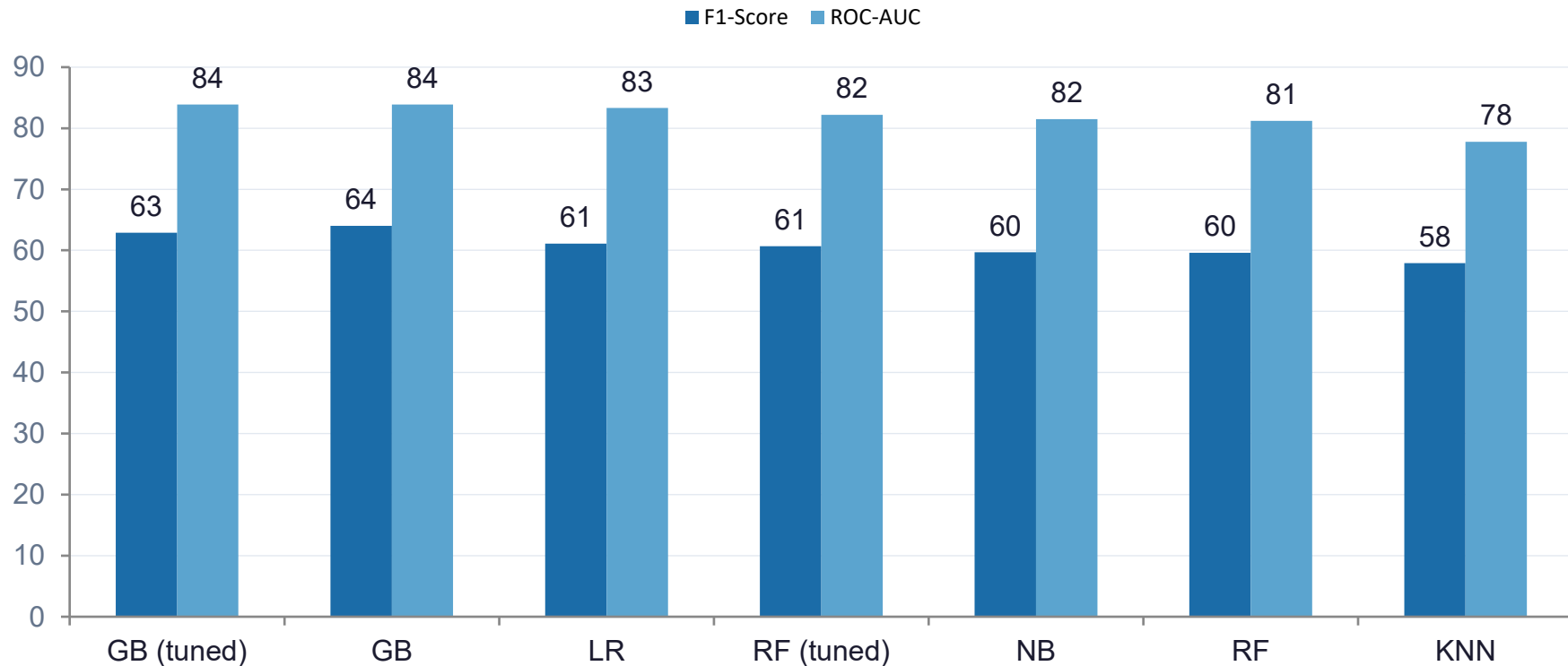
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Cloud Scaling

Move from Streamlit to FastAPI + cloud (AWS/GCP) for enterprise scale

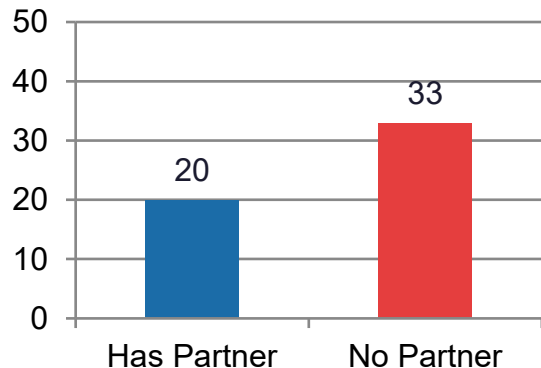
Backup — Full Model Metrics (All Models & Variants)

F1-Score vs ROC-AUC by Model (%)

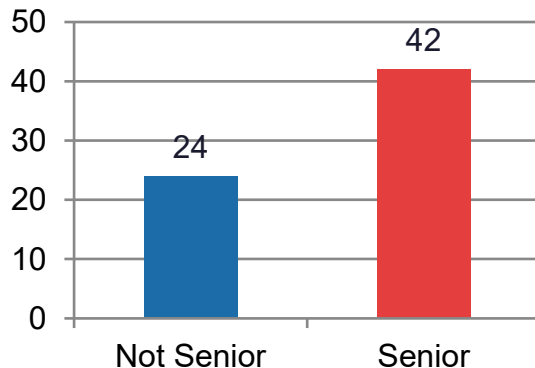


Backup — Additional EDA: Partner, Dependents & Senior Citizen

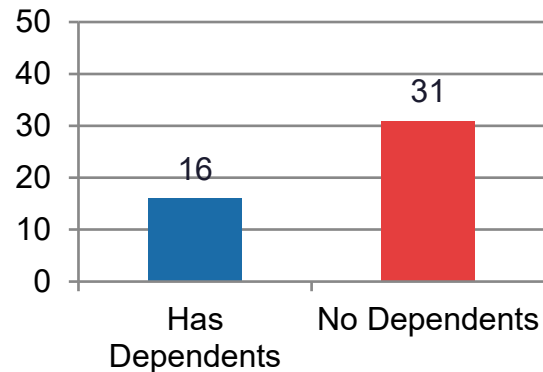
Churn by Partner (%)



Churn by Senior Citizen (%)



Churn by Dependents (%)



- Customers without a partner churn at 33% vs. 20% with a partner
- Senior citizens churn at 42% vs. 24% for non-seniors
- Customers without dependents churn at 31% vs. 16% with dependents

Backup — Full Recommendations Summary

#	Finding	Recommendation
1	Month-to-month contracts have 43% churn	Incentivize migration to longer-term contracts
2	Fiber optic churn = 42%	Improve service quality; targeted retention offers
3	Electronic check churn = 45%	Incentivize switch to auto-pay or credit card
4	No partner/dependents = higher churn	Family and partnership bundled service discounts
5	Senior citizens churn at 42%	Dedicated senior support and simplified plans
6	No value-added services = higher churn	Promote security, backup, and tech support bundles
7	Short tenure customers churn more	Improve onboarding; offer first-year loyalty bonus
8	Paperless billing users churn more (34%)	Digital-first engagement and personalized comms

Milestone 3 — Modeling Pipeline

1. Data Split



Stratified 80/20 split preserving 73/27 churn ratio

2. SMOTE Balancing



Applied only to training set to prevent data leakage on test set

3. 5 Classifiers Trained



Logistic Regression, Random Forest, Gradient Boosting, KNN, Naïve Bayes

4. 5-Fold CV



SMOTE wrapped inside imblearn Pipeline — no leakage into validation fold

5. Hyperparameter Tuning



GridSearchCV applied to top-2 baselines: Gradient Boosting & Random Forest

6. Winner Selection



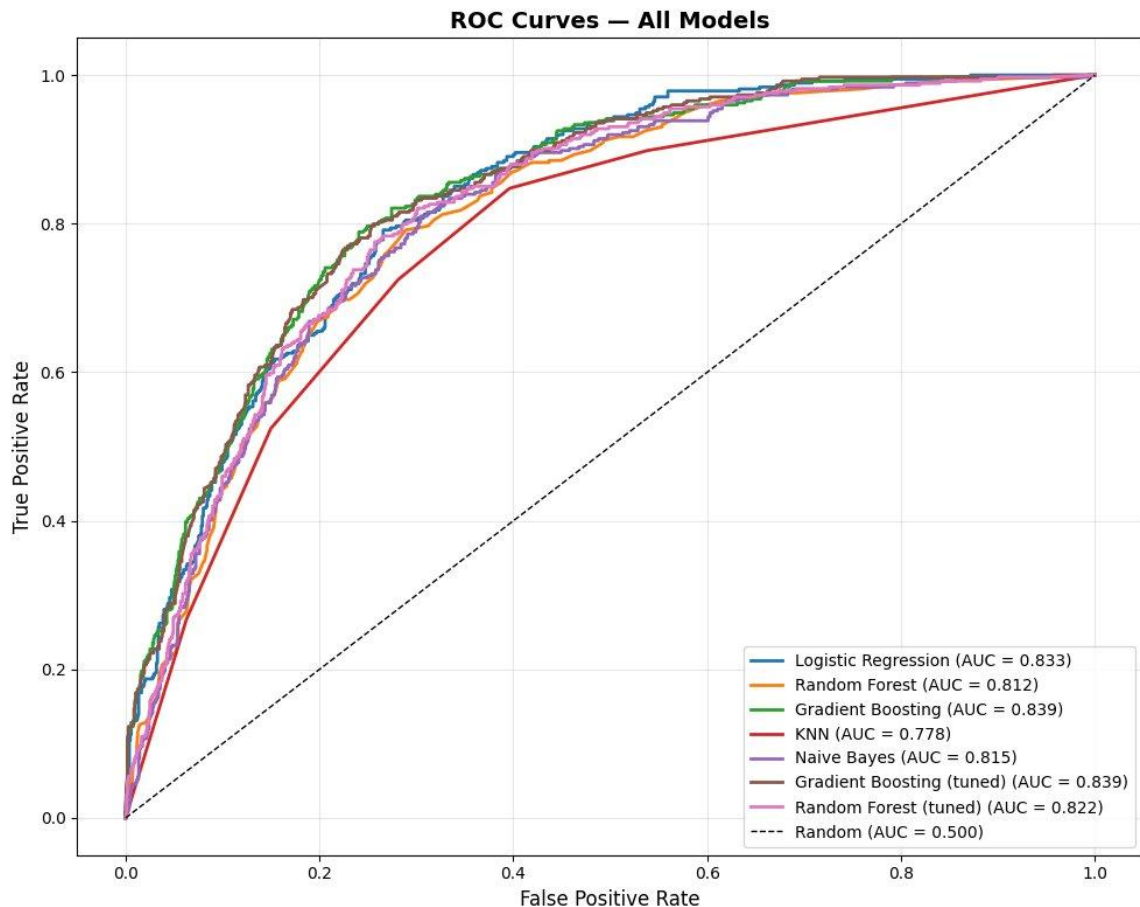
Final model chosen by F1-score — optimal for imbalanced churn detection

Milestone 3 — Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
★ Gradient Boosting (tuned)	74.7%	51.5%	80.8%	62.9%	0.839
Gradient Boosting	76.4%	53.7%	79.1%	64.0%	0.839
Logistic Regression	72.7%	49.1%	80.8%	61.1%	0.833
Random Forest (tuned)	77.0%	55.6%	66.8%	60.7%	0.822
Naïve Bayes	69.9%	46.3%	83.9%	59.7%	0.815
Random Forest	76.9%	55.6%	64.2%	59.6%	0.812
KNN	72.0%	48.2%	72.5%	57.9%	0.778

★ Winner by F1-Score and Recall — best for churn detection

Milestone 3 — Model Evaluation: ROC Curves



Top AUC Scores

Gradient Boosting (tuned)

AUC = 0.839

Logistic Regression

AUC = 0.833

Random Forest (tuned)

AUC = 0.822

Naïve Bayes

AUC = 0.815

Milestone 3 — Confusion Matrices (All Models)

